

Lessons

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Grouping Symbols and Separated Terms

In algebraic expressions and equations, terms are separated by addition and subtraction symbols. A term can consist of a number alone, a variable alone, or the product of a number and a variable: 12, y , or $3x$.

Like terms have exactly the same variables. Constants are also like terms. You’ve already learned how to simplify expressions by combining like terms. Now, you want to simplify the sides of an equation so that you can solve it.

Example 1

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Solve: $-6 + 2n = 7n + 1 - 4n$

Step 1

Combine like terms on each side of the equation.
The right side of this equation can be simplified.

$$\begin{aligned} -6 + 2n &= 7n + 1 - 4n \\ -6 + 2n &= 7n - 4n + 1 \\ -6 + 2n &= 3n + 1 \end{aligned}$$



Step 2

Isolate the variable on one side of the equation.
It doesn’t matter which side you choose.

$$\begin{aligned} -6 + 2n &= 3n + 1 \\ \text{Subtract } 2n \text{ from each side.} \\ -6 + 2n - 2n &= 3n - 2n + 1 \\ -6 &= n + 1 \\ \text{Subtract 1 from each side.} \\ -6 - 1 &= n + 1 - 1 \\ -7 &= n \end{aligned}$$



$-7 = n$ is the same as $n = -7$.

Substitute -7 for n in the original equation to check.

$$\begin{aligned} -6 + 2n &= 7n + 1 - 4n \\ -6 + 2(-7) &= 7(-7) + 1 - 4(-7) \\ -6 + (-14) &= -49 + 1 + 28 \\ -20 &= -20 \end{aligned}$$



The equation checks. The answer is $n = -7$.

Use these steps for solving equations:

1. Simplify each side of the equation.
2. Isolate the variable on one side of the equation using inverse operations.

Students often assume that if an answer is a whole number, then it must be right. They also assume that if the answer is a fraction or a decimal, it must be wrong. These are false assumptions. Many correct answers in real life are fractions or decimals.

The only way to be sure your answer is correct is to substitute your solution into the equation. Good test takers think about whether their answers make sense, and they check their work if they have enough time.

When an equation contains parentheses, you must remove them before you can solve the equation. You have already learned about the distributive property. Use the distributive property to remove parentheses and other grouping symbols.

Example 2

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Solve for m . $4 - 3m = 2m + 7(m - 8)$

As it is written, $7(m - 8)$ is a product. You need to separate the product into terms so that you can combine like terms and use inverse operations to solve the equation.

Step 1

Use the distributive property.

$$\begin{aligned} 4 - 3m &= 2m + 7(m - 8) \\ 4 - 3m &= 2m + 7m - 56 \end{aligned}$$

Step 2

Combine like terms.

$$\begin{aligned} 4 - 3m &= 2m + 7m - 56 \\ 4 - 3m &= 9m - 56 \end{aligned}$$

Step 3

Use inverse operations.

$$4 - 3m = 9m - 56$$

Add $3m$ to both sides.

$$4 = 12m - 56$$

Add 56 to both sides.

$$60 = 12m$$

Divide both sides by 12.

$$5 = m$$

The solution to the equation is $m = 5$.

Don’t be fooled when you see a subtraction sign before parentheses. To find the value of an expression like $-(x - 3)$, imagine there is a 1 before the parentheses. Then multiply the terms in the parentheses by -1 . $-(x - 3) = -1(x - 3) = -x + 3$

A fraction bar is another kind of grouping symbol. You can eliminate a fraction by multiplying both sides of the equation by the denominator of the fraction.

Example 3

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Solve for x . $\frac{3(8x-1)}{5} = x + 7$

Step 1

Use the distributive property to eliminate the parentheses in the numerator.

$$\begin{aligned} \frac{3(8x-1)}{5} &= x + 7 \\ \frac{24x-3}{5} &= x + 7 \end{aligned}$$

Step 2

The denominator is 5. Multiply both sides of the equation by 5. The 5s cancel. Use the distributive property on the right side of the equation.

$$\begin{aligned} \cancel{5} \cdot \left(\frac{24x-3}{\cancel{5}} \right) &= \cancel{5}(x + 7) \\ 24x - 3 &= 5x + 35 \end{aligned}$$

Step 3

Use inverse operations.

$$24x - 3 = 5x + 35$$

Add 3 to both sides.

$$24x = 5x + 38$$

Subtract $5x$ from both sides.

$$19x = 38$$

Divide both sides by 19.

$$x = 2$$

The solution to the equation is $x = 2$.

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Category
Lesson 3: Solving Equations

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